

Question:

A 5-year-old boy is brought to the physician by his mother because he does not “listen to her” anymore. The mother also reports that her son cannot concentrate on any tasks lasting longer than just a few minutes. Teachers at his preschool report that the patient is more active compared to other preschoolers, frequently interrupts or bothers other children, and is very forgetful. Last year the patient was expelled from another preschool for hitting his teacher and his classmates when he did not get what he wanted and for being disruptive during classes. He was born at term via vaginal delivery and has been healthy except for 3 episodes of acute otitis media at the age of 2 years. He has met all developmental milestones. His mother has major depressive disorder and his father has Graves' disease. He appears healthy and well nourished. Examination shows that the patient does not seem to listen when spoken to directly. The remainder of the examination shows no abnormalities. Which of the following is the most appropriate next step in treatment?

Answer Choices:

- A. Behavior therapy
- B. Methimazole
- C. Fluoxetine
- D. Hearing aids

Reasoning + Answer:

<think>

Finding Reasoning Paths :

- 1 . **Attention-deficit/hyperactivity disorder (ADHD)**: The symptoms described (inattentiveness , hyperactivity , impulsivity , inability to concentrate , forgetfulness , disruptive behavior , and short attention span) are classic for ADHD . ADHD is a common neurodevelopmental disorder in children and can be treated with stimulant medications like methylphenidate or amphetamines , though behavior therapy and/or educational interventions are often recommended as a first-line approach .
- 2 . **Major depressive disorder (MDD)**: The mother's history of MDD could explain some symptoms , such as inattentiveness or irritability , but the presence of hyperactivity and impulsivity makes ADHD more likely .
- 3 . **Graves ' disease (hyperthyroidism)**: Hyperthyroidism in children can cause behavior changes , but it typically presents with increased energy , weight loss , tachycardia , and other systemic symptoms , which are not described here .
- 4 . **Childhood-onset bipolar disorder**: Bipolar disorder can present with hyperactivity , impulsivity , and mood swings , but the clinical picture is less consistent with it .
- 5 . **Intellectual disability**: Inattentiveness , poor impulse control , and hyperactivity could be signs of an intellectual disability , but the normal developmental milestones and lack of other intellectual or developmental delays make this less likely .
- 6 . **Fetal alcohol syndrome**: This condition can cause behavioral abnormalities , but the absence of growth retardation , facial abnormalities , or other characteristic features makes it unlikely .
- 7 . **Behavioral abnormality**: ADHD could be a non-psychiatric condition , but the constellation of symptoms strongly suggests a psychiatric etiology .

Reasoning Process :

- 1 . **Symptom Analysis** :
 - The child exhibits inattentiveness , hyperactivity , impulsivity , forgetfulness , poor concentration , and disruptive behavior . These symptoms have lasted more than six months and interfere with daily activities and educational settings .
 - The absence of growth retardation , facial abnormalities , or other systemic symptoms makes conditions like fetal alcohol syndrome or intellectual disability less likely .
 - The mother's history of MDD could contribute to some symptoms but does not account for hyperactivity , impulsivity , and poor concentration .
- 2 . **Differential Diagnosis** :
 - **Attention-deficit/hyperactivity disorder (ADHD)**: The DSM- 5 criteria for ADHD are met , and the symptoms are consistent with the condition . ADHD is the most likely diagnosis .
 - **Major depressive disorder (MDD)**: While MDD can cause irritability and behavioral changes , it does not explain the hyperactivity , impulsivity , or poor concentration .
 - **Childhood-onset bipolar disorder**: Bipolar disorder can cause hyperactivity and mood swings , but the symptom pattern (e.g. , forgetfulness , poor concentration) and the lack of mania or hypomania make this diagnosis less likely .
 - **Fetal alcohol syndrome**: The absence of growth retardation , facial abnormalities , or other characteristic features makes this unlikely .
 - **Other conditions**: Intellectual disability , Graves ' disease , or systemic illnesses are less likely given the normal developmental milestones and lack of systemic symptoms .
- 3 . **Treatment Consideration** :
 - The first-line treatment for ADHD , particularly in a 5-year-old , is behavioral therapy and/or educational interventions to improve attention , impulse control , and social skills .
 - If behavior therapy fails or there are significant behavioral or attention-related symptoms , stimulant medications like methylphenidate or amphetamines may be considered .

Conclusion :

Behavior

1.5

1.0

0.5

0.0

−0.5

−1.0

−1.5